Chapter- 01: Setting Up a Development Environment

Chapter- 02: Strings and Regular Expressions

1. **How many data type in JavaScript's?** a. 4 b. 8 c. 12 d. 16
2. **Every JavaScript object has a built-in-**
   a. Number Method b. Boolean Method c. to String Method d. big Int Method
3. **If you want to apply formatting like commas, a currency symbol, or some other locale-specific details, You need to help of the-** a. Intl.Number Format object b. Intl.Boolean Format object c. Intl.String Format object d. Intl.Null Format object
4. **The escape sequences in JavaScript all begin with the character-** a. / b. $ c. # d. \
5. **If you want to insert an extended Unicode character how encoding you need?**
   a. 4-byte b. 8 byte c. 16-byte d. none
6. **The off-the-cuff approach is to use the-**
   a. String.toUpperCase() b. String.toLowerCase() c. a and b d. none of them
7. **If localeCompare() deems that two string match, it returns-** a. 2 b. 1 c. 4 d. 0
8. **One common problem that thwarts the trim() method is ..... excess whitespace inside a string?**
   a. adding b. removing c. a and b d. none of them
9. **Programmers use many different regular expressions to validate?**
   a. UserName b. Password c. Contact No d. Email Address

Chapter- 03: Numbers

1. Which of the following statements is true regarding the comparison of BigInt and Number values in JavaScript?
a. Always use the strict equality operator (===) for comparing BigInt and Number values.
b. The loose equality operators (== and !=) should be avoided when comparing BigInt and Number values.
c. Converting a Number to BigInt before comparison using === is recommended.
d. JSON.stringify() can be used directly on a BigInt without any issues.

2. What happens when you try to serialize a BigInt using JSON.stringify()?
a. The BigInt is silently converted to a Number b. A syntax error is thrown
c. The BigInt is converted to a string d. The BigInt is ignored

3. What is the recommended approach to test for equality between a BigInt and a Number?
a. Use the loose equality operator (==) b. Use the strict equality operator (===)
c. Explicitly convert the Number to a BigInt and use === d. Convert both values to strings and compare them

4. What is the minimum and maximum base value allowed for converting decimal numbers in JavaScript?
a. 1 and 9 b. 2 and 10 c. 2 and 36 d. 3 and 35

5. Which method is generally preferred for converting strings to numbers in JavaScript?
a. Multiplying the string by 1 b. Using the unary plus operator
c. Using the Number() function d. Using the parseFloat() function

6. Which of the following statements is true about converting numbers to strings in JavaScript?
a. It's a delicate process that can often fail b. It's a safe operation that always succeeds
c. It's only possible for whole numbers, not decimals d. It requires the use of external libraries

7. In which systems can rounding errors occur while representing real numbers?
a. Decimal notation only b. Binary notation only
c. Both decimal and binary notations d. Integers only

8. Why can't some decimal values be perfectly represented in binary?
a. Binary has fewer digits than decimal b. There are infinitely many real numbers
c. Binary uses base 2, while decimal uses base 10 d. All of the above

9. Which Math object method always rounds a fractional number down to the nearest whole number?

a. round() b. floor() c. ceil() d. abs()

10. Which method rounds a fractional number up to the next whole number?

a. round() b. floor() c. ceil() d. abs()

11. What happens when you use round() on the exact value 0.5?

a. It rounds down to 0 b. It rounds up to 1

c. It remains unchanged at 0.5 d. The behavior depends on rounding settings

12. What is a "typed array" used for in the context of Crypto.getRandomValues()?

a. To specify the desired number of random values b. To define the range of possible random values

c. To store the generated random values in a specific data type

d. To provide additional options for customizing the randomness

13. What type of numbers does Crypto.getRandomValues() generate?

a. Truly random numbers b. Cryptographically strong pseudorandom numbers

c. Regular pseudorandom numbers like Math.random() d. Depends on the browser implementation

14. What type of array does Crypto.getRandomValues() fill with random values?

a. A standard JavaScript array b. A typed array c. An object d. A string

15. Which of the following is NOT a valid typed array type for use with Crypto.getRandomValues()?

a. Uint8Array b. Uint16Array c. Float32Array d. Int32Array

Chapter- 04: Dates

1. What does the following JavaScript code do? const today = new Date();

A. Creates an empty Date object B. Initializes a Date object with the current date and time

C. Sets the system time to the current date and time D. Gets the local date and time from the user's computer

2. In JavaScript, how does one extract the current year from a Date object?

A. date.getYear() B. date.getFullYear() C. date.getYearNumber() D. date.currentYear()

3. JavaScript's Date object is perfect for handling complex date calculations. A. True B. False

4. Which statement is true about creating a Date object for a specific date?

A. Months are numbered 1-12, with January as 1 B. Days are numbered 0-30, with the 31st being 0

C. You can optionally specify hours, minutes, seconds, and milliseconds

D. The constructor accepts a string representation of the date

5. What is the advantage of using UTC methods like getUTCMonth()?

A. They are more accurate for local time B. They are preferred for calculations involving multiple time zones

C. They can handle daylight saving time changes automatically D. They are required for all date operations

6. Which of the following is the most reliable format for converting a string to a Date object?

A. "12/30/2021" B. "Dec 30, 2021" C. "2021-12-17T03:24:00Z" D. "December 30, 2021"

7. What is the recommended way to create a Date object from a string in ISO 8601 format?

A. new Date.parse("2021-12-17T03:24:00Z") B. Date.create("2021-12-17T03:24:00Z")

C. new Date("2021-12-17T03:24:00Z") D. Date.fromString("2021-12-17T03:24:00Z")

8. Which method is used to modify the day of a Date object?

A. changeDate() B. setDate() C. modifyDate() D. adjustDate()

9. How can you compare two Date objects to see which one comes first chronologically?

A. Use the < or > operators B. Use the == operator

C. Use the compare() method D. Use the isBefore() and isAfter() methods

10. Why does date1 === date2 often return false even when two Date objects represent the same date?

A. Because the === operator only checks for object reference equality, not value equality.

B. Because Date objects are inherently mutable and cannot be compared directly.

C. Because Date objects are always created with different millisecond values.

D. Because Date objects must be explicitly converted to numbers before comparison.

11. The Date object can represent dates as far back as 271,821 BCE and as far forward as 275,760 CE.

A. True B. False

12. Which method can be used to create a Date object that only represents a date, without any time information?

A. setDate(0, 0, 0) B. setHours(0, 0, 0, 0) C. stripTime() D. toDateOnly()

13. True or False: The getTime() method returns a negative number for dates in the past and a positive number for dates in the future. A. True B. False

14. What does the Math.trunc() function do when calculating the difference in days between two dates?

A. It adds 1 to the result to ensure whole days are counted B. It rounds the result to the nearest integer

C. It removes any time information from the result D. It converts the result to milliseconds

15. What options can you specify when using the Intl.DateTimeFormat object to format a date?

A. Only year, month, and day
B. Year, month, day, hour, minute, second, weekday, and era
C. User-defined formatting patterns
D. All of the above.

Chapter- 05: Arrays

1. In which method you can iterate over your array in functional way ?

a) Array.Round()
b)Array.forEach()
c) Array.loops()
d) Array.array()

2. How do you iterate through each element in an array using a for loop?

a) for(let in array)
b) for(array.forArrasy)
c) for(let element of array)
d) for(let i=0;i<array.length;++i)

3. How can you sort array of strings or numbers?

a) Array.for()method
b) Array.forEach()method
c) Array.sort()method
d) Array.find()method

4. Why Arrays can hold any type of JavaScript Object?

a)Every Array is a class
b) Every Array is an Object
c) Every Array is a value
d) None

5. How can you Merging Two Arrays into a new array?

a) Array.add()method
b) Array.pluse()method
c) Array.merge()method
d) Array.concat()method

6. How can you make a shallow copy of a portion of an existing array ?

a) Array.slice()method
b) Array.add()method
c) Array.cut()method
d) Array.delete()method

7. What is splice()method use for?

a) for add
b) for replace or delete
c) for find
d) for numbers

8. Which comparison means 3 and '3' are not considered?

a) 3/3
b) ===
c) +/+
d) =/=

9. By which method you can flatten a two-dimensional array into one-dimensional?

a) Array.flat()method
b) Array.delete()method
c) Array.oned()method
d) Array.cut()method

10. Which one is the functional array searching methods?

a) find() or findIndex()
b) where()
b) not()
d) search()

11. --------is not an array .but -like an array?

a) Set
b) List
c) Valu
c) func

12. What we should always use for adding a items to a Map object?

a) set()method
b) add()method
c) call()method
d) for()method

Chapter- 06: Functions

1. What is a significant limitation of using arrow functions in JavaScript when compared to traditional function declarations?

a. They cannot be assigned to variables.
b. They do not support default parameters.
c. They have a different way of binding the `this` keyword.
d. They cannot be used as callback functions.

2. When providing a function as an argument to another function, what potential issue may arise when using an anonymous function declared inline?

a. It leads to a syntax error.
b. It results in a less readable stack trace during debugging.
c. It cannot be passed as a parameter to certain methods.
d. It causes the function to be inaccessible outside its scope.

3. In JavaScript, why might using the `arguments` object be considered less favorable than using the rest parameter for functions that accept an unlimited number of arguments?

a. The `arguments` object requires explicit declaration.
b. The `arguments` object is not iterable.
c. The `arguments` object is not an actual array.
d. The `arguments` object cannot be used in arrow functions.

4. When using default parameters in JavaScript functions, what is a potential drawback of setting default values for multiple parameters?

a. It violates the DRY (Don't Repeat Yourself) principle.
b. It leads to a higher risk of naming conflicts.
c. It makes all parameters after the first one mandatory.
d. It may result in less readable code.

5. What happens if you try to use the `yield` keyword in an arrow function in JavaScript?

a. It works as expected.
b. It throws a syntax error.
c. It is interpreted as a regular function.
d. It creates a generator function.

6. In the context of passing functions as arguments, what is a potential advantage of providing a reference to a function declared elsewhere in your code?

a. It ensures the function is always inlined for better performance.
b. It allows the function to be reused in different parts of the application.

c. It improves the readability of the function call. d. It enforces stricter scoping rules.

7. Consider the function declaration:
`function processArray (callback, ...values)`. What does the use of the rest parameter (`...values`) signify in this context?

a. It collects all arguments after the first one into an array.
b. It ensures that `callback` is always provided as the first argument.
c. It restricts the number of arguments that can be passed to the function.
d. It specifies a default array to be used when no values are passed.

8. In the context of arrow functions, why might they be less suitable for use as object methods or constructors in JavaScript?

a. They lack the ability to bind the `this` keyword. b. They cannot have return statements.
c. They result in longer stack traces during debugging. d. They have limited support for default parameters.

9. When declaring an inline function using arrow syntax, why might it be considered good practice to include a function name?

a. It enhances the function's execution speed. b. It ensures compatibility with older JavaScript versions.
c. It improves code readability and aids in debugging. d. It prevents the function from being inlined.

10. What is the primary advantage of using arrow functions with a single statement over traditional function declarations when passing them as arguments?

a. They enforce stricter scoping rules. b. They allow for more concise code.
c. They provide better support for default parameters. d. They automatically handle errors more efficiently.

11. When using named function parameters with object literals, what is a potential disadvantage related to preventing property naming mistakes?

a. It leads to a syntax error if property names are incorrect. b. It introduces a performance overhead.
c. It may not catch all property naming mistakes. d. It makes the code less readable.

12. What is the purpose of using a closure in JavaScript, as explained in the context?

A. To terminate the execution of a function. B. To encapsulate variables and functions, preserving their state.
C. To create random numbers within a specific range. D. To simplify the creation of generator functions.

13. How can closures be used to store and preserve data across function calls?

A. By using global variables. B. By passing data as parameters to each function call.
C. By creating a stateful function package. D. By utilizing the `setTimeout` function.

14. In the context of closures, what does the term "state" refer to?

A. The condition of the JavaScript engine. B. The current execution stack.
C. The values of variables in the outer function. D. The speed of function execution.

15. Why is using global variables for storing data discouraged in JavaScript?

A. Global variables lead to naming collisions. B. Global variables improve code readability.
C. Global variables simplify debugging. D. Global variables have faster access times.

16. What does the outer function in the closure structure return?

A. The inner function. B. The parameter values. C. Global variables. D. The outer function.

17. What is the key advantage of using closures to store data across several function calls?

A. Simplicity in code structure. B. Enhanced code reusability.
C. Faster execution speed. D. Automatic garbage collection.

18. What is the primary purpose of using generator functions in JavaScript?

A. To simplify the creation of closures. B. To create functions with a fixed set of parameters.
C. To provide multiple values on-demand with pausing execution. D. To replace the need for global variables.

19. What does the term "closure" refer to in JavaScript?

A. The process of closing a JavaScript file after execution.
B. A function that encapsulates its surrounding state, allowing it to persist.
C. The ability to close browser windows using JavaScript. D. The end of the execution of a JavaScript program.

20. In the context of generator functions, what does the `yield` keyword do?

A. It terminates the generator function. B. It pauses the execution of the generator function until resumed.
C. It skips the current iteration in the generator function. D. It returns the final result of the generator function.

21. What is the primary purpose of using a generator function in JavaScript?

A. To create functions that run indefinitely. B. To handle time-consuming data creation operations.
C. To replace the need for loops in JavaScript. D. To generate random numbers in a specific range.

22. What is the primary benefit of using partial application in JavaScript?

A. Reducing the need for closures. B. Simplifying the creation of generators.
C. Eliminating the use of functions with multiple parameters.
D. Reducing redundancy by creating specialized function versions.

Chapter- 07: Objects

1. What does the instanceof operator do in JavaScript?

a) Checks if an object is empty
b) Tests if an object is an instance of a specific type
c) Determines if an object has a certain property
d) Iterates over all the properties of an object

2. How can you test if an object is not an instance of a specific type using the instanceof operator?

a) if (mysteryObject !instanceof Date)
b) if (!mysteryObject instanceof Date)
c) if (!(mysteryObject instanceof Date))
d) if (mysteryObject instanceof !Date)

3. What is the purpose of using the typeof operator when testing a variable that might hold one of the primitive data types?

a) To check if the variable is an instance of a specific type
b) To determine if the variable is not an instance of a specific type
c) To get one of nine predefined string values indicating the type
d) To test if the variable is empty

4. In JavaScript, what is the purpose of using an object literal?

a) To determine if an object has a certain property
b) To test if an object is not an instance of a specific type
c) To group several variables together to create a basic data package
d) To iterate over all the properties of an object

5. How can you check if an object has a given property in JavaScript?

a) Using the Object.getOwnPropertyNames() method
b) Using the Object.values() method
c) Using the in operator
d) Using the Reflect.ownKeys() method

6. What is the purpose of using Object.keys() when iterating over all the properties of an object?

a) To retrieve all the property values of an object
b) To count the number of properties in an object
c) To get an array with the property names of an object
d) To create a new object with computed property names

7. How can you test if an object is empty in JavaScript?

a) Using the Object.values() method
b) Using the isEmpty() method from libraries like Underscore or Lodash
c) Checking if the object is equal to a predefined empty object
d) Using Object.keys() and checking if the length is 0

8. What does the for...in loop do in JavaScript?

a) Iterates over all the properties of an object
b) Tests if an object is an instance of a specific type
c) Checks if an object has a given property
d) Determines if an object is empty

9. What is the result of merging two objects using the spread operator?

a) A new object with properties from both objects
b) The first object overwrites properties of the second object
c) The second object overwrites properties of the first object
d) Merging objects is not possible in JavaScript

10. How can you explicitly customize the way a property is defined in an object?

a) Using the spread operator
b) Using the Object.assign() method
c) Using Object.defineProperty() method
d) Using the Object.freeze() method

11. Which method is used to freeze an object against any changes?

a) Object.preventExtensions()
b) Object.seal()
c) Object.freeze()
d) Object.lock()

12. What does the Proxy class allow you to do?

a) Create new objects
b) Intercept and change actions on an object
c) Merge two objects

d) Clone an object

13. How can you create an exact copy of a custom object?

a) Using Object.assign()
b) Using the spread operator (...)
c) Using Object.freeze()
d) Using Object.defineProperty()

14. Why might you need a deep copy of an object?

a) To make the object immutable
b) To ensure all properties are enumerable
c) To copy only the top-level properties
d) To copy nested objects and their properties

15. What type can be used to create absolutely unique object property keys?

a) String
b) Number
c) Symbol
d) Boolean

16. What advantage does using Symbol offer when creating enums?

a) Guaranteed uniqueness
b) Simplicity in syntax
c) Better performance
d) Compatibility with JSON

Chapter- 08: Classes

1. JavaScript an object-oriented Programming Language.

a. True
b. False

2. Object Oriented Programming languages usually revolve around concepts like-

a. Classes
b. Interfaces
c. Inheritance
d. all of above

3. **You want to create a _____ template for custom objects.**

a. usable
b. reusable

4. JavaScript Class is a _____ function

a. Constructor
b. Generator
c. Auto invoke
d. Named

5. JavaScript all methods are attached to that Functions _____

a. Parameter
b. variable function
c. sub-Class
d. Prototype

6. Constructor function are always named _____

a. Class
b. Constructor
c. parameter type
d. none

7. When you write a constructor, you use this to create new _____ on the current object.

a. Public Field
b. Class
c. Loval variable
d. Global Variable

8. newPerson.dateOfBirth= new date (2010,10,10);
console.log(newPerson.dateOfBirth);

a. This Change is allowed
b. This change causes an error

9. To perform validation before updating a_____

a. Field
b. Class
c. Constructor
d. method

10. To expose a single property of an object that's stored in a _____

a. Field
b. Class
c. Constructor
d. method

11. To use some kind of lazy initialization, which only creates or calculates a _____when its needed.

a. property value
b. parameter
c. Parameter Value
d. method

12. The default implementation of toString() for all objects displays the unhelpful text.

a. True
b. False

Chapter- 09: Asynchronous Programming

1. When you pass a function as an argument, remember not to use parenthesis?

a. setTimeout (myFunction, 3000)
b. setTimeout (myFunction (), 3000)
c. setTimeout(myFunction, 3000 ())
d. setTimeout(myFunction,() {} 3000)

2. Do you use the async and await key-words in anonymous function!

a. True
b. False

3. Which method is called presented repeatedly?

a. setTime ();
b. setTimeOut();
c. setTimeout()
d. setTimeFunction();

4. What should happen in the case of an error you call……function?

a. Time.catch()
b. Promise.catch()
c. Error.catch()
d. Cnsole.catch()

5. "A Promise Object is not a result but it's shown a placeholder for result" -this statement is

a. False
b. True

6. How many ways to stop work in async function ……..?

a. One Ways
b. Two Ways
c. Three Ways
d. None Ways

7. Which function is use to standard message-passing of your work……..?

a. postMessage({"message"})
b. primesMessage({"message"})
c. showMessage({"message"})
d. useMessage({"message"})

8. Which event is commonly used to perform asynchronous operations in the browser environment?

a. click
b. load
c. change

d. submit

9. Which method is used to catch errors in a Promise?

a. catchError
b. error
c. catch
d. onError

10. What is the main purpose of asynchronous programming in JavaScript?

a. To make code execution sequential
b. To handle user inputs
c. To manage asynchronous tasks and avoid blocking the main thread
d. To reduce code complexity

11. The WebSocket object contains the following events:

a. onclose, onmessage, and onopen.
b. onclose, onmessage, onerror, and onopen.
c. onclick, onmessage, onerror, and onopen.
d. onclose, onmouseover, onerror, and onopen.

12. You want to chain several asynchronous calls together. Which promise method will you use?

a. the file method on the promise object.
b. the keep method on the promise object.
c. the pipe method on the promise object.
d. There is no way for this.

13. When you need to create a promise object whose state is resolved, then you will use-----

a. $.when()
b. $.get()
c. $.param()
d. Both 'a' and 'b'

14. The …….. Object is a wrapper for the promise object. The deferred object provides control of the promise object, which is read-only.

a. Deferred
b. Prepared
c. Preferred
d. Differed

15. The …… method can be used to get the current state of the promise object.

a. State
b. Estate
c. currentStat
d. currentState

16. Which method creates a resolved promise object?

a. resolve()
b. done()
c. $.when()
d. $.Deferred()

17. Which method do you call on the deferred object to indicate a change in progress?

a. progress
b. notify
c. done
d. resolve

18. Web workers are useful for execution of a script file in a background task.

a. True
b. False

19. Communication to and from the web worker is accomplished by using the ……. method.

a. postMessage
b. postMassage
c. sendMessage
d. sentMessage

20. The WebSocket protocol is a web technology that provides full-duplex communications over a single……connection.

a. HTTP
b. FTP
c. TCP
d. All of the above

21. The WebSocket object contains methods to send data and close the connection.

a. True
b. False

22. …… code has one sequential execution path, and when the long-running code is executing, nothing else can be accomplished.

a. Semi-Synchronous
b. Asynchronous
c. Shrinking
d. Synchronous

23. The *promise* (also known as a *future* or *deferred*) object provides a mechanism to schedule work to be done on a value that might not yet be computed.

a. True
b. False

24. The subscription functions execute a maximum of one time.

a. True
b. False

25. The name of all functions that perform asynchronous calls should end with Async, and these functions should always return a promise object.

a. True
b. False

26. Chained and parallel asynchronous operations automatically pass failure and progress to the end of the chain.

a. True
b. False

27. Which method chains asynchronous operations?

a. pipe()
b. done()
c. resolve()
d. always()

28. Which method do you call to subscribe to the successful completion of an asynchronous operation?

a. done
b. always
c. complete
d. success

29. The worker can send messages to the spawning task by posting messages to an event handler specified by the creator and messages can be any object that can be serialized.

a. True
b. False

30. The worker thread blocks the user interface thread, so the UI remains fast and fluid.

a. True
b. False

31. Web workers' state is ……. from the webpage.

a. Isolated
b. Non-isolated
c. Grouped
d. United

32. The web worker does not have access to DOM elements.

a. True
b. False

33. In the web worker code, you want the asynchronous code to update the DOM. Which method can you use?

a. updateDom()
b. workerUpdate()
c. dom()
d. You cannot update the DOM from within the asynchronous code

34. Which object can be used when the web worker and the creator need to reference the same object?

a. Mutex
b. Semaphore
c. Closure
d. The web worker and the creator cannot reference the same object

35. Do the WebSocket protocol and WebSocket API provide native support for load balancing in a web farm?

a. Yes
b. No

Chapter- 10: Errors and Testing

1. Which property of the error object provides a human-readable description of the error?

A. subtype
B. name
C. stack
D. message

2. Why might using nonstandard properties like "description" and "lineNumber" in error handling be problematic?

A. They are not supported in any browser.
B. They only work in specific browsers.
C. They provide inaccurate information.
D. They can cause syntax errors.

3. What information does the stack property of the error object provide?

A. Browser-specific details about the error
B. Currently open functions on the stack, in order
C. Human-readable description of the error
D. The line number where the error occurred

4. In error handling, when is the finally block executed?

A. Before the try block
B. After the try block if no errors occurred, or after the catch block if an error was caught
C. Only if an error occurs in the try block
D. Only if an error occurs in the catch block

5. The catch block in a try...catch structure receives an error object that contains information about the error, including its name, message, and stack.

A. True
B. False

6. Which error type occurs when a numeric value is outside of its allowed range?

A. ReferenceError
B. SyntaxError
C. RangeError
D. TypeError

7. JavaScript allows multiple catch blocks for different error types within a single try block.

A. True
B. False

8. The catch block in a try...catch structure can handle both standard JavaScript error types and custom error objects.

A. True
B. False

9. How many catch blocks are allowed for a single try block in JavaScript?

A. One
B. Two
C. Unlimited
D. Three

10. What is the recommended approach for handling errors in asynchronous operations using the older XMLHttpRequest API?

A. Wrap the asynchronous operation in a try...catch block
B. Attach an error event handler to the XMLHttpRequest object
C. Use the Promise.catch() method
D. Utilize the window.unhandledrejection event

11. In the async and await model, where should the error-handling code be placed for a promise-based operation?

A. Inside the async function before the promise
B. As a separate callback function
C. Inside the then() method
D. In the catch section attached to the promise

12. How can you catch unhandled errors in your JavaScript code and create a diagnostic log?

A. Use a try...catch block in the main application thread
B. Attach an event handler to window.onerror
C. Utilize Promise.catch() for asynchronous errors
D. Implement logging in the Fetch API

13. How can you create an instance of the Error object in JavaScript?

A. new Error('Custom error message')
B. Error('Custom error message')

C. createError('Custom error message')
D. throw Error('Custom error message')

14. Which properties are available in the Error object?

A. Only message
B. message, name, and type
C. message, name, and stack
D. error, description, and stacktrace

15. When throwing an error for a numeric value outside an acceptable range, which error subtype is suitable to use?

A. ReferenceError
B. SyntaxError
C. TypeError
D. RangeError

16. Why is setting the name property of a custom error class important?

A. It improves the error message readability
B. It is required for the error to be caught in a try...catch block
C. It persists even if the error object is serialized to JSON
D. It automatically captures the stack trace

17. What is the purpose of using Error.captureStackTrace() in the constructor of a custom error class?

A. To capture the current state of the call stack
B. To suppress the stack trace information
C. To prevent the error from being thrown
D. To modify the behavior of the throw statement

18. The Error.captureStackTrace() method is optional and does not affect the behavior of a custom error class.

A. True
B. False

19. Which tool is recommended for writing unit tests in JavaScript according to the provided solution?

A. Mocha
B. Jasmine
C. Jest
D. Karma

20. What file extension does Jest expect for test files?

A. .test
B. .test.js
C. .spec
D. .spec.js

21. Which Jest matcher function is used to check for exact equality in the test examples?

A. toEqual()
B. toBe()
C. toBeCloseTo()
D. toBeGreaterThan()

22. What does the "Statements" percentage in a code coverage report represent?

A. Percentage of code functions tested
B. Percentage of code branches reached
C. Percentage of code statements executed during tests
D. Percentage of uncovered lines in the code

23. Achieving 100% test coverage is universally agreed upon as the best goal, and anything below that is considered inadequate.

A. True
B. False

Chapter- 11: Browser Tools

1) What action does the 'Step into' option perform in JavaScript debugging?

a) Skips current function
b) Execute a function
c) continue the code in full
d) Steps into a function to debug it further

2) Which statement about breakpoints is true?

a) They provide less control than console.log
b) They don't display values in scope
c) They offer less flexibility than console.log
d) They provide greater control and offer more flexibility

3) What is the shortcut to open Chrome's Developer Tools?

a) Command-Option-C
b) Control-Alt-D
c) Control-Shift-C
d) Alt-Shift-D

4) From which tab in Chrome's DevTools you can set the breakpoint?

a) Source
b) Console
c) Element
d) Performance

5) Which developer tools helps to Performance analysis and look for bottlenecks?

a) Source
b) Console
c) Element
d) Performance

6) What will be shown if we click the reload button in performance tab?

a) Performance metrics related to a new page load
b) Code execution details
c) Network request information
d) Browser settings

7) What does the Chrome Performance tooling do?

a) Simulates a fast internet connection
b) Analyzes and presents the browser's rendering process
c) Records audio in the browser
d) Checks for broken links in the webpage

8) How to spot performance issues that users will encounter but not apparent to developer?

a) Increase the machine's CPU speed
b) Use a high-speed internet connection
c) Clear the browser cache
d) Throttle the CPU or internet connection in browser performance tools

9) What is the purpose of the Chrome Developer Tool's Coverage feature?

a) Debugging JavaScript
b) Identifying unused JavaScript
c) Analyzing runtime performance
d) Setting breakpoints

10) How to stop recording the results in Chrome's Developer's Tool?

a) Press F5
b) Click the Stop Instrumenting Coverage And Show Results button
c) Close the browser tab
d) Use the "coverage-stop" command in the console

11) What is Lighthouse?

a) A web browser
b) Chrome's Developer Tool
c) A open source tool
d) A code editor

12) Lighthouse Generates reports and recommendations for which area?

a) Performance and Accessibility
b) SEO and Progressive Web application
c) Best Practices
d) All of the above

13) How can you access Lighthouse in Chrome's Developer Tools?

a) Click on the Lighthouse extension icon
b) Use the "lighthouse" command in the console
c) Click the DevTools Lighthouse tab
d) Lighthouse is not available in Chrome's Developer Tools

14) Which browser's developer tools does Lighthouse integrate with?

a) Firefox
b) Safari
c) Google Chrome
d) Internet Explorer

15) Which device types can you select when using Lighthouse?

a) Desktop only
b) Mobile only
c) Both A & B
d) Lighthouse doesn't support device type selection

Chapter- 12: Working with HTML

1. **What do you mean by DOM?**
   a. Document Original Model
   b. Document Object Model
   c. Document Objectors Model
   d. Document Object ore Model
2. **The DOM is a programming interface for _____ documents that defines the logical structure of documents and the way a document is accessed and manipulated?**
   a. HTML
   b. XML
   c. HTML and XML
   d. XAML
3. **Dutton is a name that could refer to many things.**
   f3 True
   g3 False
4. **How many selectors are there in JavaScript?**
   a. 2
   b. 3
   c. 4
   d. 5
5. **What is API?**
   a. Application Program Interface
   b. Applicate Programming Interface
   c. Application Programming Interface
   d. Application Program and Interface
6. **How many selectors are there in API?**
   a. 2
   b. 3
   c. 4
   d. 5
7. **What is WAI?**
   a. Web Accessibility Initiative
   b. Web Accessibility Initiative
   c. Web Accessibility Initiative
   d. Web Accessibility Initiative
8. **What is ARIA?**
   a. Accessible Rich Internet Applications wed
   b. Accessible Rich Internet Applications
   c. Accessible Riches Internet Applications
   d. Accessible Rich Internet web Applications

9. Which is not a reserved word JavaScript? a) interface b) short c) program d) throws

10. When you want to send form data to the server to update data in the database, which HTTP method should you specify of the <*form*> element? a. POST b. GET c. CONNECT d. SEND

11. A web service provides communications between----

a. A web server and a file b. A web server and a client c. A developer and a client d. Both 'a' and 'b'

12. You want to use jQuery to make an AJAX call that will use the HTTP POST method. Which jQuery method can you use?

a. $.post() b. $.getJSON() c. $.get() d. $.param()

13. This '102' describes the categories of return status codes for--------

a. Success b. Client error c. Server error d. Informational message

14. How do you write "Hello World" in an alert box?

a) alertBox="Hello World" b) msgBox("Hello World") c) alert("Hello World") d) alertBox("Hello World")

15. How do you create a function? a) function myFunction() b) function:myFunction() c) function=myFunction()

16. Html5 does not support more than one form submission element. This is--------- a. True b. False

17. Which sign does jQuery use as a shortcut for jQuery? A. the % sign B. the ? Sign C. the $ sign

18. Which jQuery method is used to hide selected elements?

A. hidden() B. hide() C. visible(false) D. display(none)

19. Which is/are true for form submission elements?

a. *<button type='submit'>* b. *<input type='submit'>* c. *<input type='image'>* d. All of the above

20. If you want to stop your jQuery for a few milliseconds, which function do you use?

a) stop() b) pause() c) slowdown() d) delay()

21. Which function do you reference in HTML form data?

a) value() b) formData() c) val() d) None of the above

22. Which input type defines a slider control? a. controls b. slider x c. search d. range

23. Which HTML5 element is used to display a scalar measurement within a known range?

a. <measure> b. <gauge> c. <meter> d. <range>

24. The web server sends a request **to the web browser, by using HTTP protocol. This is----** a. True b. False

25. How do you write a conditional statement for executing some code if "i" is equal to 5?

a) if i=5 then b) if (i==5) c) if i==5 then d) if i=5

26. What is the primary object used to make an asynchronous AJAX call that returns a JSON result?

a. XMLHttpRequest b. AjaxRequest c. JsonResponse d. AjaxAsync

27. This '502' describes the categories of return status codes for----------

a. Success b. Client error c. Server error d. Informational message

28. The primary object that makes an AJAX call is----

a. XMLHttpResponse b. XMLHTTPRequest c. XMLHttpRequest d. XMLHtpsRequest

29. With jQuery, look at the following selector: $("div"). What does it select? A. All div elements B. The first div element

30. Which jQuery method is used to perform an asynchronous HTTP request?

A. jQuery.ajaxAsync() B. jQuery.ajax() C. jQuery.ajaxSetup()

31. You are creating a webpage that will collect data and send it to a server. Which element must you implement to be able to send data to the server? a. *<nav>* b. ***<form>*** c. *<textarea>* d. * *

32. When you retrieve a webpage from the server that contains Customer information, you want to be able to specify the id of the Customer within the URL. What can you use to provide the id?

a. the PostBack b. the QueryString c. the domain name d. a label

33. You have serialized the form data that has an *<input type = "email">* element. You want to deserialize the data to see the email address as entered. Which function can you use?

a. decodeURIComponent() b. deserialize() c. submit() d. deserializeEmail()

34. Java Script entities start with _____and end with_____

a. Semicolon, colon b. Semicolon, Ampersand c. Ampersand, colon d. Ampersand, semicolon.

35. In HTML5, onblur and onfocus are: a. Event attributes b. Style attributes c. HTML elements

36. Which method to remove an event listener? a. .on b. .off c. .remove d. none

37. Which one is true for "form"?

a. Form can sent data back to the server for processing b. Form can't sent data back to the server for processing

38. Which is or are true for web browser return status code?

a. 5xx Server error b. 2xx Success c. 3xx Redirect d. 4xx Client error e. c & d

39. The method that provides AJAX functionality within jQuery is named. A. ajax B. easyAJAX C. jQueryajax D. jQajax

40. Is it possible to declare a variable in Java Script along its type? a. Yes b. No

41. The speed options can be applied to which jQuery functions? a) css and ajax b) show and fadeIn c) toggleCss

Chapter- 13: Fetching Remote Data

1. What is the primary purpose of the Fetch API?

A. To create interactive animations on web pages.

B. To style and format web page elements.

C. To retrieve and process data from remote servers asynchronously.
D. To manipulate the Document Object Model

2. Which of the following methods can be used to parse the JSON data from a successful response?

A. parseJSON() B. json() C. readData() D. extractContent()

3. Which method is used to prevent the default form submission behavior in JavaScript?

A. stopSubmit() B. preventDefault() C. cancelSubmit() D. haltSubmit()

4. What is the primary purpose of the FormData object in JavaScript?

A. To create and format HTML forms.
B. To create key/value pairs of form data for transmission.
C. To validate user input in forms.
D. To style form elements.

5. Which method is used to append a new field to a FormData object?

A. addField() B. appendField() C. append() D. insert()

6. What is the correct way to send a POST request with a FormData object using fetch?

A. fetch(url, { method: 'post', body: formData })
B. fetch(url, { method: 'post', data: formData })
C. fetch(url, { method: 'post', formData })
D. fetch(url, formData)

7. Which method is used to retrieve a specific value from a FormData object by its key?

A. getValue() B. get() C. getItem() D. find()

8. Which method is used to check if a FormData object contains a value with a given key?

A. contains() B. has() C. includes() D. exists()

9. What is the main advantage of using XMLHttpRequest (XHR) over fetch when working with older browsers?

A. XHR has broader browser support, including older versions of Internet Explorer.
B. XHR offers more advanced features for form submission.
C. XHR is generally faster and more efficient than fetch.
D. XHR is easier to use and learn than fetch.

10. Which method is used to create a JavaScript object from a JSON-formatted string?

A. JSON.create() B. JSON.parse() C. JSON.convert() D. JSON.transform()

11. Which method is used to convert a JavaScript object into a JSON-formatted string?

A. JSON.stringify() B. JSON.serialize() C. JSON.format() D. JSON.encode()

12. Which of the following data types are valid in JSON?

A. Strings, numbers, booleans, arrays, objects, null
B. Strings, numbers, booleans, arrays, objects, functions
C. Strings, numbers, booleans, arrays, objects, dates
D. Strings, numbers, booleans, arrays, objects, undefined

13. How are boolean values represented in JSON?

A. As the keywords true and false.
B. As the numbers 1 and 0.
C. As the strings "true" and "false".
D. As the symbols true and false.

14. Which of the following is a valid response type for a fetch() request?

A. text B. json C. blob D. All of the above

15. What is CORS?

A. A technique for sharing resources between different domains.
B. A security measure that prevents cross-site scripting attacks.
C. A way to compress data to reduce network bandwidth.
D. A protocol for real-time communication between a client and server.

16. What is WebSockets?

A. A protocol for full-duplex communication between a client and server.
B. A technique for long polling a server.
C. A way to cache data on the client side.
D. A security measure that protects against man-in-the-middle attacks.

17. What is long polling?

A. A technique for maintaining a connection to a server by repeatedly sending requests.
B. A way to compress data to reduce network bandwidth.
C. A security measure that prevents cross-site scripting attacks.
D. A protocol for real-time communication between a client and server.

Chapter- 14: Data Persistence

1. How is the max-age option used in cookies?

a) It sets the domain for which the cookie is accessible.
b) It sets the length of time that the cookie is valid.
c) It sets the path at which the cookie is accessible.
d) It specifies whether the cookie should be transmitted over HTTPS.

2. What does the encodeURIComponent function do in the context of cookies?

a) It encodes strings and removes unnecessary characters.
b) It decodes strings and removes unnecessary characters.
c) It sets the expiration date for a cookie.
d) It specifies the path at which the cookie is accessible.

3.How long does a sessionStorage last?

a) As long as the browser is open
b) Until the user logs out
c) Across multiple browser sessions
d) Until the user closes the tab or window

4. Which function is used to remove an individual item from sessionStorage?

a) removeItem
b) clear
c) deleteItem
d) eraseItem

5. What is the key difference between sessionStorage and localStorage?

a) localStorage lasts only for the current session.
b) localStorage is accessible across multiple browser sessions.
c) localStorage is specific to the tab or window.
d) localStorage has a smaller storage capacity.

6. Which event handler is invoked when a new version of the database is opened in IndexedDB?

a) onsuccess
b) onerror
c) onupgradeneeded
d) onclick

7. What is the purpose of creating indexes in the object store of IndexedDB?

a) To limit the size of the database
b) To speed up data retrieval
c) To enforce uniqueness constraints
d) To clear stored data

8. What does the await database.getAll('reader') method do in the getData function?

a) Deletes all data in the 'reader' object store
b) Retrieves all data from the 'reader' object store
c) Adds new data to the 'reader' object store
d) Updates existing data in the 'reader' object store

9. Which method is used to open a database and returns a promise in the IDB library?

a) openDatabase
b) created
c) openDB
d) initDB

10. In JavaScript, localStorage persists data across browser sessions, while sessionStorage stores data for a single session only.

a) True
b) False

11. The IndexedDB API provides a synchronous way to work with client-side data storage.

a) True
b) False

Chapter- 15: Working with Media

1. What is the primary reason for including CDATA sections around JavaScript within an SVG file?

A. To improve performance by caching the script.
B. To prevent XML parsers from misinterpreting script characters.
C. To provide type information for the browser.
D. To enable cross-origin resource sharing.

2. Which JavaScript library is specifically designed for working with SVG animations?

A. jQuery
B. React
C. Raphaël
D. Bootstrap

3. Which of the following DOM methods is NOT directly applicable to SVG elements?

A. document.getElementById()
B. document.createElement()
C. document.write()
D. element.setAttribute()

4. Which method is used to access SVG elements embedded directly in a web page?

A. object.contentDocument
B. object.getSVGDocument()
C. document.getElementById()
D. svgdoc.getElementById()

5. What is the primary difference between accessing SVG in HTML vs. an external SVG file?

A. The browser compatibility for SVG features.
B. The need for error handling.
C. The method used to retrieve the SVG document object.
D. The ability to modify SVG attributes.

6. Which of the following is a valid way to access an SVG element with the ID "square" within an external SVG file?

A. const square = svgdoc.getElementsByTagName("square")[0];

B. const square = svgdoc.getElementById("square
C. const square = document.querySelector("#square");
D. const square = object.contentDocument.getElementById("square");

7. Which method is used to assign a function to the onclick event handler of an SVG element?

A. svgdoc.addEventListener("click", function() { ... });
B. element.onclick = function() { ... };
C. svgdoc.addEventHandler("click", function() { ... });
D. element.setAttribute("onclick", "function() { ... }");

8. The video element has a built-in stop method.

A. True
B. False

9. Which D3 method is used to create placeholders for missing elements based on provided data?

A. append()
B. select()
C. data()
D. scale()

10. Which video format is supported by nearly all modern browsers?

A. H.264 (.mp4)
B. VP8 (.webm)
C. Ogg Theora (.ogv)
D. All of the above

11. What event is triggered periodically during video playback?

A. timeupdate
B. progress
C. playupdate
D. timestamp

12. What attribute is used to display built-in controls for media elements?

A. controls
B. show_controls
C. toolbar
D. ui

Chapter- 16: Writing Web Application

1. What is the purpose of the bundle.js file created by Webpack?

a) It contains the entire npm registry
b) It is a configuration file for Webpack
c) It is a minified output of the JavaScript modules
d) It is used for hot module reloading

2. What does code splitting in the context of Webpack mean?

a) Breaking the code into smaller modules
b) Generating multiple bundles for different browsers
c) Eliminating dead or unused code
d) Loading only the necessary code for a specific page

3. What is tree shaking in the context of Webpack?

a) Reducing the size of bundled JavaScript files
b) Eliminating dead or unused code
c) Generating multiple bundles for different browsers
d) Minifying JavaScript files

4. How can you enable lazy loading of JavaScript modules in Webpack?

a) Set lazyLoad: true in the webpack.config.js file
b) Add a lazyLoad script in the package.json file
c) Use dynamic import statements with import()
d) Install the webpack-lazy-load package

5. What does the sideEffects field in the package.json file control in Webpack?

a) Code minification
b) Lazy loading
c) Dead code elimination
d) Handling files with special behaviors

6. Which of the following is a key role of the service worker in a Progressive Web Application (PWA)?

a) Rendering the HTML content
b) Directly accessing the DOM
c) Running in the background and enabling offline functionality
d) Managing the application's display style

7. Why is the service worker script registration placed at the bottom of the HTML page?

a) To prioritize the service worker over other scripts
b) To improve performance by loading it last
c) To ensure it runs before other scripts
d) To avoid interference with other scripts in the head section

8. What is the purpose of the fetch event in the service worker script?

a) Installing the service worker
b) Activating the service worker

c) Handling fetch events and providing guidelines for loading resources
d) Deleting old cache items

9. What is a limitation of service workers that developers need to be aware of?

a) They cannot access the DOM directly
b) They only work in private browsing mode
c) They require a specific version of the web app manifest
d) They must be written in a specific programming language

10. What does the File API provide in the context of file handling in the browser?

a) A list of files to upload
b) Information about a specific file, including filename, size, and MIME type
c) An object to asynchronously upload files for client-side access
d) All of the above

11. What is the purpose of the FileReader object in the File API?

a) To provide a list of files to upload
b) To provide information about a specific file, including filename, size, and MIME type
c) To asynchronously upload files for client-side access
d) To track the progress of a file upload

Chapter- 17: Node Basics

1. Which statements are true about Node Version Manager (NVM)?

a) To manage multiple versions of Node on your development machine
b) To create a Node application that can respond to a very basic browser request
c) To ensure version matching across environments, including deployment
d) All of the above

2. What is the primary purpose of Node's REPL?

a) To manage multiple Node.js versions
b) To create Node web servers
c) To interactively run Node code snippets
d) To compile Node code into machine code

3. How do you start Node's REPL from the command line?

a) node app.js
b) npm start
c) node
d) repl

4. Which of the following objects are accessible from Node's global object?

a) console
b) Buffer
c) require()
d) All of the above

5. What function is used to customize the prompt in REPL?

a) setPrompt()
b) customizePrompt()
c) REPL.prompt()
d) None of the above

6. What is the default prompt in REPL? a) > b) $ c) # d) node>

7. What keystroke combination exits REPL?

a) ctrl+c
b) ctrl+d
c) .exit
d) All of the above

8. Which function is used to read a file's contents asynchronously?

a) readFile()
b) readFileSync()
c) getFile()
d) getContent()

9. Which function is used to write data to a file, overwriting its existing contents?

a) writeFile()
b) appendFile()
c) createFile()
d) writeToFile()

10. Which function is used to append text to the end of an existing file?

a) writeFile()
b) appendFile()
c) appendToFile()
d) createFile()

11. What is the difference between a buffer and a string in Node.js?

a) A buffer can handle binary data, while a string cannot.
b) A string can handle binary data, while a buffer cannot.
c) A buffer is always encoded as UTF-8, while a string can have any encoding.
d) A string is always encoded as UTF-8, while a buffer can have any encoding.

12. What problem does the callback hell phenomenon refer to?

a) The difficulty of debugging asynchronous code with nested callbacks

b) The performance overhead of using asynchronous functions
c) The complexity of managing file permissions in Node.js
d) The challenge of handling errors in file I/O operations

13. What is the recommended way to handle file I/O in Node.js to avoid callback hell?
a) Use synchronous functions like readFileSync() and writeFileSync()
b) Use promises or async/await to manage asynchronous operations more cleanly
c) Use a third-party library that simplifies file I/O
d) Write all file I/O code in a single, large function

14. What function is used to close the Readline interface? a) close() b) end() c) finish() d) terminate()

15. Which variable contains the path of the currently executing script?
a) __filename b) __dirname c) __path d) __currentScript

16. Which variable contains the directory of the currently executing script?
a) __filename b) __dirname c) __path d) __currentDirectory

17. What is the scope of the __dirname and __filename variables?
a) Global scope b) Module scope c) Function scope d) Block scope

18. How would you create a new directory named "temp" within the current script's directory?
a) fs.mkdirSync(__dirname + "/temp");
b) fs.mkdirSync(__filename + "/temp");
c) fs.mkdirSync(path.join(__dirname, "temp"));
d) fs.mkdirSync(path.join(__filename, "temp"));

Chapter- 18: Node Modules

1. How can you search for Node.js modules directly using the npm command-line interface?
A. npm query module-name
B. npm find module-name
C. npm search module-name
D. npm locate module-name

2. What is the purpose of searching for a module using the npm website?
A. To execute the module's code
B. To download and install the module
C. To discover detailed information about the module
D. To create a new module

3. How can you refine the results of a module search using multiple terms in the npm command-line interface?
A. Use the --filter option
B. Add more keywords after the search command
C. Use the --refine option
D. Use the -m flag

4. In Node.js, what keyword is used to define what's exported from a library/module?
A. import B. expose C. export D. exports

5. What is the purpose of the require() function in Node.js module system?
A. To define exports in a module
B. To include a module in the application
C. To declare functions within a module
D. To load external libraries

6. How can you export a function directly using the module object in Node.js?
A. module.defineFunction = myFunction
B. module.createFunction = myFunction
C. module.exports = myFunction
D. module.function = myFunction

7. What file extension is commonly used for CommonJS modules in Node.js?
A. .js B. .cjs C. .mjs D. .node

8. Which keyword is used for exports in CommonJS modules?
A. import B. expose C. export D. exports

9. What field in the package.json file specifies the main entry point of the application for CommonJS modules?
A. entry B. main C. source D. start

10. The README.md file often contains Markdown syntax for better readability and is used to provide documentation for the project. A. True B. False

11. The npm link command is used to publish a Node module to the npm registry. A. True B. False

12.What field in the webpack.config.js file specifies the name of the library that will be used in ECMAScript, CommonJS, and AMD module environments? A. module B. libraryName C. library D. moduleName

13. What testing framework is used in the solution for unit testing the module?
A. Mocha B. Jasmine C. Jest D. Chai

14. Which command is used to run the Jest tests in the provided solution?
A. npm start B. npm run test C. npm run jest D. npm run unit-test

15. What is the purpose of the expect function in Jest unit tests?
A. To execute the test suite
B. To define the test cases
C. To create mock functions
D. To assert and verify test outcomes

1."Modularizing ensures that external functionality is created in such a way that it isn't dependent on other external functionality" a concept known as___ ?
A. Tight coupling.
B. Code entanglement.
C. Modular interdependence.
D. Loose coupling.

2.Node's default module system is based on____?
A. AMD (Asynchronous M7odule Definition).
B. ES6 Modules.
C. CommonJS.
D. UMD (Universal Module Definition.)

3. How many constructs are there in Node's default module system?
A. One construct.
B. Two constructs.
C. Three constructs.
D. Four constructs.

4. which of the following is NOT Node's default module system constructs?
A. exports.
B. require().
C. module.
D. define().

5. How many key fields are there in package.json configuration?
A. One.
B. Two.
C. Three.
D. Four.

6. What does the "type" key field in the package.json specify?
A. It indicates the main entry point of the application.
B. It defines the exported paths of the modules.
C. The library is using ECMAScript module syntax or CommonJS.
D. It points to the CommonJS file for the library.

7. What does the "main" key field in the package.json specify?
A. The main entry point of the application.
B. It defines the exported paths of the modules.
C. The library is using ECMAScript module syntax or CommonJS.
D. It points to the CommonJS file for the library.

8. What does the "exports" key field in the package.json specify?
A. The main entry point of the application.
B. It defines the exported paths of the modules.
C. The library is using ECMAScript module syntax or CommonJS.
D. It points to the CommonJS file for the library.

9. How can you configure the package.json file to avoid using the .cjs and .mjs file extensions in a project using ECMAScript module syntax?
A. Set "main" to "./index.mjs" for the main entry point.
B. Change "exports" to include ".": "./index.js" and "./module": "./wrapper.js".
C. Define "type" as "commonjs" to exclusively use CommonJS.
D. Use ".cjs" and ".mjs" file extensions to ensure compatibilit.

10. What command is used to remove a module from the development dependencies in the package.json file?
A. $ npm remove express.
B. $ npm remove jest.
C. $ npm install jest --save-dev.

D. $ npm install express –save.

11. How can you add a development dependency, such as a testing framework like Jest, to your package.json file and update it simultaneously?
A. $ npm install jest.
B. $ npm install jest --save-dev.
C. $ npm install express –save.
D. $ npm install express.

12. Which field in the package.json file is responsible for specifying a description of what the Node module is and does?
A. main.
B. author.
C. description.
D. keywords .answer.

13. Which markup language is commonly used for creating README files, emphasizing readability and simplicity?
A. HTML
B. Markdown
C. XML
D. JSON

14. What does the term "UMD" stand for in the context of JavaScript module environments?
A. Universal Module Design
B. Unified Module Deployment.
C. Universal Module Definition.
D. Ultimate Module Deployment.

15. What does "CLI" stand for?
A. Common Language Interface.
B. Command Line Interface.
C. Component Level Integration.
D. Centralized Library Invocation.

16. Which file contains the configuration for Webpack and the Webpack command-line interface (CLI) as development dependencies?
A. webpackrc.
B. webpack.config.js.
C. package-lock.json.
D. package.json.

17. If you want to expose a library as a variable in the UMD build, what should you set the libraryTarget field to in Webpack?
A. umd
B. var
C. UMD
D. variable

18. What does the npm run build (or yarn run build with Yarn) command do in the provided example?
A. It installs necessary dependencies for the project.
B. It starts a development server for the library.
C. It generates the Webpack bundle for the library.
D. It runs tests for the library.

1. What does the process.env object in Node.js provide access to?

A) Local storage data B) Environment variables
C) User session information D) File system properties

2. How can you set an environment variable when running a Node.js application?

A) By using a JSON file B) By declaring it in the script
C) By specifying it ahead of the node command D) By storing it in a database

3. Which npm package is recommended for managing environment variables stored in a .env file?

A) express B) nodemon C) dotenv D) mongoose

4.How do you initialize the dotenv package in your Node.js application?

A) import { config } from 'dotenv' B) require('dotenv').initialize()
C) require('dotenv').config() D) dotenv.start()

5.What is a good practice when dealing with default values for environment variables?

A) Hardcode the default values in your code B) Use an || operator to specify a default value
C) Always require a value to be set in the environment D) Store default values in a separate .env.defaults file

6.Where should you not commit your .env file?

A) To a public repository B) To a private repository
C) To version control D) To a secure, encrypted database

7. What Node.js feature can be used to manage "Callback Hell"?

A) Callback functions B) Event listeners C) Async/await syntax D) Promises

8. Since which version of Node.js is the async/await syntax available? A) 6.0 B) 7.0 C) 8.0 D) 10.0

9. What does the promisify utility in Node.js convert a function to return?

A) An event emitter B) A callback C) A promise D) An observable

10. What is a primary feature of Node.js design regarding code execution?

A) Synchronous execution B) Asynchronous, or nonblocking, execution
C) Sequential execution D) Parallel execution

11.How can errors be handled when using async/await syntax?

A) Using if/else statements B) Using the .catch() method on promises
C) Using try/catch blocks D) Using error events

12. What is the benefit of using async/await over traditional callback functions?

A) Increases the execution speed of the code B) Makes the code more readable and easier to maintain
C) Eliminates the need for external libraries D) Allows for synchronous code execution

13. Which utility is used to convert a callbackbased function to return a promise, simplifying asynchronous operations?

A) async B) await C) promisify D) callbackify

14. What method is used to launch a given process with optional commandline arguments in Node.js?

A) exec() B) spawn() C) fork() D) execFile()

15. When using the spawn() method, which event is used to handle the standard output (stdout) from the child process?

A) close B) exit C) data D) error

16. What property of the process object in Node.js returns an array containing commandline arguments passed to the program? A) process.env B) process.argv C) process.cmd D) process.args

17. Which module simplifies working with commandline arguments in Node.js by parsing them into an object?

A) commander B) process C) child_process D) yargs

18. What is the output format of the argv property when using Yargs to parse commandline arguments?

A) String B) Array C) Object D) Boolean

19. What module is recommended for building commandlinedriven applications in Node.js?

A) http B) commander C) express D) yargs

20. Which option allows you to define a version for a commandline utility using Commander?

A) .version() B) .help() C) .option() D) .parse()

21. How can you specify commandline options/arguments with Commander?

A) Using .arg() B) Using .command() C) Using .option() D) Using .parse()

22. What does the npm link command do for a Node.js commandline utility?

A) Compiles the utility into binary format B) Makes the utility globally available on the user's system
C) Uploads the utility to the npm registry D) Links the utility to the Node.js executable

23. To add a commandline utility to the package.json file, under which property should it be included?

A) "scripts" B) "dependencies" C) "bin" D) "main"

24. How can you access the values of commandline arguments passed to a Node.js script?

A) process.env.arguments B) process.argv C) console.log(arguments) D) require('args')

25. Which of the following is a correct way to make use of commandline arguments in a Node.js application?

A) Parsing process.argv manually B) Using the exec method of the child_process module
C) Using the commander or yargs module for easier parsing D) All of the above

26. When using the spawn method to execute a command in a child process, what type of object does the data event handler receive? A) A string B) A Buffer C) An array D) An object

27. What does the chmod a+x command do to a file?

A) Changes the file to be readable by all users
B) Changes the file to be writable by all users
C) Makes the file executable by all users
D) Deletes the file for all users

28. What tool is recommended for ensuring a Node.js application restarts if it shuts down unexpectedly in a production environment? A) nodemon B) npm C) pm2 D) nodecron

29. Which utility automatically restarts your Node.js application when code changes are detected during development?

A) npm B) pm2 C) nodecron D) nodemon

30. Which module allows you to schedule tasks using GNU crontab syntax in Node.js?

A) pm2 B) nodecron C) exec D) nodemon

31. In the crontab syntax 0 0 * * *, what does this schedule represent?

A) A task that runs every minute
B) A task that runs at the start of every month
C) A task that runs at midnight every day
D) A task that runs every hour

32. Which type of testing is essential for determining how an application performs under stress?

A) Unit testing B) Performance testing C) Integration testing D) Manual testing

33. What tool can be used for load testing to simulate many concurrent clients accessing a Node.js application?

A) Selenium B) loadtest C) Thor D) Both B and C

34. Which Node.js module is used to execute shell commands or run other programs?

A) http B) fs C) child_process D) path

35. Which child_process method is suitable for spawning a new process with a shell?

A) fork() B) exec() C) spawn() D) execFile()

36. What is a key advantage of using spawn() over exec() in Node.js?

A) spawn() can run commands synchronously.
B) spawn() is used to execute JavaScript files only.
C) spawn() is suitable for processes that return large amounts of data.
D) spawn() can only run shell commands, not executable files.

37. What does the fork() method in the child_process module do?

A) Executes a shell command
B) Creates a duplicate of the current process
C) Spawns a new Node.js process
D) Splits a process into two concurrent processes

38. Which flag is used with pm2 to watch for file changes and restart the application automatically?

A) autorestart B) watch C) livereload D) autorefresh

39. What does the nodecron syntax */5 * * * * represent?

A) A task that runs every 5 hours
B) A task that runs at the 5th minute of every hour
C) A task that runs every 5 days
D) A task that runs every 5 minutes

40. For load testing a WebSocket application, which tool is specifically designed to test WebSocket connections?

A) loadtest B) Selenium C) Thor D) ab (ApacheBench)

41. In Node.js, which method would you use to run a nonNode.js executable file without spawning a shell?

A) exec() B) spawn() C) execFile() D) fork()

42. How can you ensure a Node.js application runs at a specific time in a specific timezone using nodecron?

A) By setting the timezone option in the schedule method
B) By modifying the server's timezone settings
C) By using the exec() method with a time condition
D) By manually adjusting the cron syntax for the timezone difference

Chapter- 20: Remote Data

1. **AJAX** is

a. Async JavaScript and XML
b. Asynchronous JavaScript and XHTML
c. Asynchronous JavaScript and XML
d. Asyness JavaScript and XML

2. **One of the most common ways to achieve this is through AJAX.**
a. **True**
b. False

3. **JSONP is a _____ that allows cross-domain requests.**
a. **Technique**
b. Model
c. Roll
d. Object

4. **CORS** is
a. Cross-Design Resource Sharing.
b. Cross-Design Review Sharing.
c. Cross-Domain Resource Sharing.
d. Cross-Domain Review Sharing.

5. **Create a server side script on domain that acts as a proxy. JavaScript code sends the data to server, which then forwards it to the remote server.**
a. **True**
b. False

6. **GPS receivers with _____.**
a. R-band radio processor
b. **L-band radio processor**
c. S-band radio processor
d. T-band radio processor

7. **Carrier phase receivers, offering GPS position accuracy of _____ cm with differential correction.**
a. 20 to 40
b. 10 to 40
c. **10 to 30**
d. 20 to 30

8. **Self-contained GPS receivers, also known as 'GPS mice'.**
a. True
b. **False.**

9. **Node-fetch is only one common and useful library for fetching remote data.**
a. True.
b. **False.**

10. **A common use case is to download data when ____ is not provided.**
a. a XML
b. an UPI
c. **an API**
d. a GPS

11. **URL** is
a. Unit Remote Locator
b. Unity Reverse Location
c. **Uniform Resource Locator**
d. Uniform Remote Location

12. **XML is a software and hardware independent tool for storing and transporting data.**
a. **True**
b. False

Chapter- 21: Building Web Applications with Express

1. What is one characteristic of Express?
A. Heavyweight and complex
B. Limited to specific types of web applications
C. Lightweight and minimal
D. Suitable only for front-end development

2. Express is often compared to which Python web framework?
A. Django
B. Flask
C. Pyramid
D. Tornado

3. Which templating engine is a popular choice for Express applications?
A. EJS
B. Handlebars
C. Pug
D. Jade

4. Which directory contains the core of the Express application?
A. /bin
B. /public
C. /routes
D. /views

5. Which middleware is responsible for serving static files in an Express application?
A. express.static()
B. express.json()
C. express.urlencoded()
D. cookieParser()

6. Which method in Express is used to handle request data?
A. app.get()
B. app.post()
C. app.put()
D. app.delete()

7. How can you enable multiple HTTP methods for a specific route in Express?
A. By using separate route handlers for each method
B. By defining a custom middleware for each method
C. By chaining the methods together using .route()
D. By specifying the methods in the route path

8. What is OAuth primarily used for?
A. Encrypting data
B. Authentication
C. Hashing passwords
D. Authorization

9. How does OAuth differ from traditional username/password authentication?
A. OAuth requires fewer steps for authentication
B. OAuth involves obtaining temporary tokens

C. OAuth uses a digital signature for authentication
D. OAuth doesn't require the user's password to be stored
10. How does OAuth handle user authorization?
A. By storing user credentials in the application
B. By redirecting the user to the authorization server
C. By encrypting user data before transmission
D. By generating random access tokens

11. What does the `oauth_token' parameter in the authorization URL represent?
A. The user's access token
B. The application's client ID
C. The temporary request token
D. The authorization code

12. Why is it beneficial to use Node modules for tasks like generating formatted data?
A. They are easy to implement
B. They are thoroughly tested and maintained
C. They are only available for specific data formats
D. They are less secure than manual methods

13. What does REST stand for in the context of web development?
A. Representational Server Technology
B. Remote Execution System Transfer
C. Representational State Transfer
D. Real-time Exchange System Transfer

14. What is the purpose of using Express in building a RESTful API with Node.js?
A. To handle database queries
B. To simplify routing and request handling
C. To encrypt data transmission
D. To generate HTML templates

15. What is the purpose of GraphQL type definitions?
A. To define the structure of the database
B. To specify the routes for API endpoints
C. To enforce a strict plan for the API
D. To generate documentation for the API